

Physical Chemistry of Materials

Lecture 08

Grand Canonical Ensemble



Intramolecular Interactions

Interaction type	Distance dependence of potential energy	Typical energy/ (kJ mol ⁻¹)	Comment
Ion-ion	$1/r$	250	Only between ions
Hydrogen bond		20	Occurs in X-H...Y, where X, Y = N, O, or F
Ion-dipole	$1/r^2$	15	
Dipole-dipole	$1/r^3$	2	Between stationary polar molecules
	$1/r^6$	0.3	Between rotating polar molecules
London (dispersion)	$1/r^6$	2	Between all types of molecules and ions

(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th Ed., Table 14B.1)



Phase Transitions

- Phase transitions: **ordered** $\leftrightarrow$ **disordered**
- e.g. liquid-gas transition, magnetic transition
- Competition between **energy** and **entropy**
- **Interactions** are required!

(Monatomic) Gas Models

- Ideal gas

- Energy $E = \sum_i (p_i^2 / 2m)$
- Partition function

$$Z = \frac{1}{N! h^{3N}} \iint e^{-\sum_i p_i^2 / 2mk_B T} d^N \vec{r} d^N \vec{p} = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2}$$

- Real gas

- Energy $E = \sum_i (p_i^2 / 2m) + \sum_{i < j} E_{\text{pot}}(r_{ij})$

Real-Gas Partition Function

$$E = \sum_i (p_i^2 / 2m) + \sum_{i < j} E_{\text{pot}}(r_{ij})$$

$$Z = \frac{1}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2} \int e^{-\sum_{i < j} E_{\text{pot}}(r_{ij}) / k_B T} d^N \vec{r}$$

$$Z = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2} \left(1 + \frac{N^2}{2V} I \right)$$

where

$$I = \int (e^{-E_{\text{pot}}(r) / k_B T} - 1) d\vec{r}$$

Evaluation of I

- **Sutherland potential:** hard sphere + dispersion force

$$E_{\text{pot}}(r) = \begin{cases} +\infty, & r < r_0 \\ -\epsilon_0 \left(\frac{r_0}{r}\right)^6, & r \geq r_0 \end{cases}$$

$$I = -\frac{4}{3}\pi r_0^3 \left(1 - \frac{\epsilon_0}{k_B T}\right)$$

By calculating $p = -(\partial A / \partial V)_{T,n}$,

one can obtain **the van der Waals equation of state**

(Thermodynamic) Ensemble

Idealized group of a large number of virtual copies
(sometimes infinitely many) of a system

Microcanonical ensemble

Canonical ensemble

Grand canonical ensemble

⋮

Grand Canonical Ensemble

Thermodynamic ensemble of an **open system**

- Conserved quantities: μ, V, T
- Microstates may have different energies
- All microstates are NOT equally probable
- Constrained optimization of entropy!

Review: Canonical Ensemble

$$S/k_B = -\sum_i p_i \ln p_i$$

$$\sum_i p_i - 1 = 0$$

We can introduce one more constraint: $\sum_i p_i E_i = U$

As the constraint implies $\sum_i p_i E_i - U = 0$,

$$\frac{\partial [S/k_B - \lambda(\sum_i p_i - 1) - \beta(\sum_i p_i E_i - U)]}{\partial p_i} = 0$$

$$p_i = e^{-1-\lambda-\beta E_i}$$

Particle Number Conservation

$$S/k_B = -\sum_i p_i \ln p_i$$

- Constraint 1: $\sum_i p_i - 1 = 0$
- Constraint 2: $\sum_i p_i E_i - U = 0$
- Additional constraint: $\sum_i n_i N_i = N_{\text{total}}$, or

$$\sum_i p_i N_i - \frac{N_{\text{total}}}{\mathcal{N}} = 0$$

Applying Lagrange's multiplier method,

$$p_i = e^{-1-\lambda-\beta E_i-\alpha N_i}$$

Grand Partition Function

$$p_i = e^{-1-\lambda-\beta E_i-\alpha N_i}$$

From the normalization constraint,

$$\sum_i p_i = e^{-1-\lambda} \sum_i e^{-\beta E_i - \alpha N_i} = 1$$

Define the **grand partition function** $\mathcal{Z}$:

$$\mathcal{Z} = e^{1+\lambda} = \sum_i \frac{e^{-\beta E_i - \alpha N_i}}{\text{Gibbs factor}}$$

Back to Entropy

$$p_i = e^{-\beta E_i - \alpha N_i} / \mathcal{Z}$$

$$S/k_B = - \sum_i p_i \ln p_i = - \sum_i p_i \ln \left(\frac{e^{-\beta E_i - \alpha N_i}}{\mathcal{Z}} \right)$$

$$S/k_B = - \sum_i p_i (-\beta E_i - \alpha N_i - \ln \mathcal{Z})$$

$$S/k_B = \beta \sum_i p_i E_i + \alpha \sum_i p_i N_i + \ln \mathcal{Z}$$

$$\therefore S/k_B = \beta U + \alpha \langle N \rangle + \ln \mathcal{Z}$$

Bridge to Thermodynamics

$$S = k_B \beta U + k_B \alpha \langle N \rangle + k_B \ln \mathcal{Z}$$

$$S = k_B \beta U + N_A k_B \alpha n + k_B \ln \mathcal{Z}$$

From thermodynamics, we know

$$\left(\frac{\partial S}{\partial U} \right)_{V,n} = \frac{1}{T}, \quad \left(\frac{\partial S}{\partial n} \right)_{U,V} = -\frac{\mu}{T}$$

Hence,

$$\beta = \frac{1}{k_B T}, \quad \alpha = -\frac{\mu}{N_A k_B T} = -\frac{\tilde{\mu}}{k_B T}$$

Grand Potential

$$S = \frac{U}{T} - \frac{\mu n}{T} + k_B \ln \mathcal{Z}$$

$$-k_B T \ln \mathcal{Z} = U - TS - \mu n = A - \mu n$$

(Legendre transform of A)

Define the **grand potential** $\phi = -k_B T \ln \mathcal{Z}$;

$$d\phi = dA - d(\mu n) = -SdT - pdV - nd\mu$$

$$\therefore \phi = \phi(T, V, \mu)$$

Derived Functions

$$\phi = -k_{\text{B}}T \ln Z$$

Using $d\phi = -SdT - pdV + nd\mu$,

$$S = -\left(\frac{\partial \phi}{\partial T}\right)_{V,\mu}, \quad p = -\left(\frac{\partial \phi}{\partial V}\right)_{T,\mu}, \quad n = \left(\frac{\partial \phi}{\partial \mu}\right)_{T,V}$$

Summary

For a grand canonical ensemble, the probability to find the system in a microstate i (with energy E_i and particle number N_i) is

$$p_i = \frac{e^{-\beta E_i - \alpha N_i}}{\mathcal{Z}} = \frac{e^{-E_i/k_B T + \tilde{\mu} N_i/k_B T}}{\mathcal{Z}}$$

where the grand partition function $\mathcal{Z} = \sum_i e^{-\beta E_i - \alpha N_i}$

Also, the grand potential is given by

$$\phi = -k_B T \ln \mathcal{Z}$$

Average Energy

$$U = \langle E \rangle = \sum_i p_i E_i = \sum_i \frac{E_i e^{-\beta E_i - \alpha N_i}}{Z}$$

However,

$$\frac{\partial \ln Z}{\partial \beta} = \frac{1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{1}{Z} \sum_i E_i e^{-\beta E_i - \alpha N_i}$$

Hence,

$$U = -\frac{\partial \ln Z}{\partial \beta}$$

Average Particle Number

$$\langle N \rangle = \sum_i p_i N_i = \sum_i \frac{N_i e^{-\beta E_i - \alpha N_i}}{\mathcal{Z}}$$

However,

$$\frac{\partial \ln \mathcal{Z}}{\partial \alpha} = \frac{1}{\mathcal{Z}} \frac{\partial \mathcal{Z}}{\partial \alpha} = -\frac{1}{\mathcal{Z}} \sum_i N_i e^{-\beta E_i - \alpha N_i}$$

Hence,

$$\langle N \rangle = -\frac{\partial \ln \mathcal{Z}}{\partial \alpha}$$

Rearrangement

$$\mathcal{Z} = \sum_i e^{-\beta E_i - \alpha N_i}$$

By grouping by the particle number, we can rearrange the summation as follows:

$$\mathcal{Z} = \sum_{N=0}^{\infty} e^{-\alpha N} \left(\sum_s e^{-\beta E_s(N)} \right) = \sum_N \left(\underbrace{e^{\mu/RT}}_{\text{fugacity}} \right)^N Z(N)$$

Calculation of $\mathcal{Z}$

$$\mathcal{Z} = \sum_N (e^{\mu/RT})^N Z(N)$$

- Distinguishable particles: $Z(N) = z^N$

$$\mathcal{Z} = \sum_N (e^{\mu/RT})^N z^N = \sum_N (ze^{\mu/RT})^N$$

- Indistinguishable particles: $Z(N) = \frac{1}{N!} z^N$

$$\mathcal{Z} = \sum_N (e^{\mu/RT})^N \frac{z^N}{N!} = \sum_N \frac{1}{N!} (ze^{\mu/RT})^N = \exp(ze^{\mu/RT})$$

Example: 3D Ideal Gas

$$z = V \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2}$$

$$\mathcal{Z} = \exp(z e^{\mu/RT}) = \exp \left[V \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2} e^{\mu/RT} \right]$$

Hence,

$$\phi = -k_B T \ln \mathcal{Z} = -k_B T V \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2} e^{\mu/RT}$$

Example: 3D Ideal Gas

$$\phi = -k_{\text{B}}TV \left(\frac{2\pi mk_{\text{B}}T}{h^2} \right)^{3/2} e^{\mu/RT}$$

$$n = \left(\frac{\partial \phi}{\partial \mu} \right)_{T,V} = -k_{\text{B}}TV \left(\frac{2\pi mk_{\text{B}}T}{h^2} \right)^{3/2} \frac{e^{\mu/RT}}{RT} = \frac{\phi}{RT}$$

$$p = - \left(\frac{\partial \phi}{\partial V} \right)_{T,\mu} = -k_{\text{B}}T \left(\frac{2\pi mk_{\text{B}}T}{h^2} \right)^{3/2} e^{\mu/RT} = \frac{\phi}{V}$$

$$\therefore \phi = pV = nRT$$

Quantum Particles

We only know the number of particles in microstate i

Consider two particles in three single-particle microstates:

- Distinguishable particles: $W = 3 \times 3 = 9$
- Indistinguishable particles: $W = 9/2! = 4.5$
- Quantum particles: for (n_A, n_B, n_C) ,
 $(1, 1, 0), (1, 0, 1), (0, 1, 1), (2, 0, 0), (0, 2, 0), (0, 0, 2) \rightarrow W = 6$

The Gibbs correction factor $1/N!$ is not 100% accurate

Bosons and Fermions

- **Bosons:** particles that **can share** a one-particle microstate with another of the same species
 - e.g. photon, helium-4 nucleus, ...
- **Fermions:** particles that **cannot share** a one-particle microstate with another of the same species
 - e.g. electron, proton, neutron, helium-3 nucleus ...
 - **Pauli exclusion principle:** two identical fermions cannot occupy the same state

Average Occupation Number

For a specific microstate with energy ϵ ,

- Classical particles (Maxwell-Boltzmann distribution)

$$n_{\text{MB}}(\epsilon) = e^{-(\epsilon - \tilde{\mu})/k_{\text{B}}T}$$

- Fermions (Fermi-Dirac distribution)

$$n_{\text{FD}}(\epsilon) = \frac{1}{e^{(\epsilon - \tilde{\mu})/k_{\text{B}}T} + 1}$$

- Bosons (Bose-Einstein distribution)

$$n_{\text{BE}}(\epsilon) = \frac{1}{e^{(\epsilon - \tilde{\mu})/k_{\text{B}}T} - 1}$$

(Distinguishable) Classical Particles

Probability of finding a single particle in state with energy ϵ :

$$p = \frac{e^{-\epsilon/k_B T}}{Z}$$

The average occupation number is

$$n_{\text{MB}}(\epsilon) = Np = \frac{N}{Z} e^{-\epsilon/k_B T}$$

However, we derived $\mu = -RT \ln(Z/N) = N_A \tilde{\mu}$;

$$\therefore n_{\text{MB}}(\epsilon) = e^{\tilde{\mu}/k_B T} e^{-\epsilon/k_B T} = e^{-(\epsilon - \tilde{\mu})/k_B T}$$

Grand Partition Function

$$\mathcal{Z} = \sum_i e^{-\beta E_i - \alpha N_i}$$

For a specific microstate with energy ϵ , i only depends on the number of particles n and $E_n = n\epsilon$:

$$\mathcal{Z} = \sum_n e^{-\beta n\epsilon - \alpha n} = \sum_n e^{-n(\epsilon - \tilde{\mu})/k_B T}$$

Fermions

$$\mathcal{Z} = \sum_n e^{-n(\epsilon - \tilde{\mu})/k_B T}$$

For a specific microstate, only $n = 0$ or $n = 1$ is possible:

$$\mathcal{Z} = 1 + e^{-(\epsilon - \tilde{\mu})/k_B T}$$

The average occupation number is

$$n_{\text{FD}}(\epsilon) = \sum_{n=0,1} n p_n = \frac{e^{-(\epsilon - \tilde{\mu})/k_B T}}{1 + e^{-(\epsilon - \tilde{\mu})/k_B T}} = \frac{1}{e^{(\epsilon - \tilde{\mu})/k_B T} + 1}$$

Bosons

The average occupation number is

$$n_{\text{BE}}(\epsilon) = \sum_{n=0}^{\infty} n p_n, \text{ where } p_n = e^{-n(\epsilon - \tilde{\mu})/k_{\text{B}}T} / \mathcal{Z}$$

Let $x = (\epsilon - \tilde{\mu})/k_{\text{B}}T$;

$$\mathcal{Z} = \sum_n e^{-n(\epsilon - \tilde{\mu})/k_{\text{B}}T} = \sum_n e^{-nx}$$

$$\frac{\partial(\ln \mathcal{Z})}{\partial x} = \frac{1}{\mathcal{Z}} \frac{\partial \mathcal{Z}}{\partial x} = -\frac{1}{\mathcal{Z}} \sum_n n e^{-nx} = -n_{\text{BE}}(\epsilon)$$

Bosons

$$\mathcal{Z} = \sum_n e^{-n(\epsilon - \tilde{\mu})/k_B T} = \sum_n e^{-nx}$$

For a specific microstate, $n = 0, 1, 2, \dots$ is possible:

$$\mathcal{Z} = 1 + e^{-x} + e^{-2x} + \dots$$

Using the closed form of the geometric series,

$$\mathcal{Z} = \frac{1}{1 - e^{-x}}$$

Bosons

$$\mathcal{Z} = \frac{1}{1 - e^{-x}}$$
$$n_{\text{BE}}(\epsilon) = -\frac{\partial(\ln \mathcal{Z})}{\partial x} = \frac{\partial}{\partial x} \ln(1 - e^{-x}) = \frac{e^{-x}}{1 - e^{-x}}$$

Hence,

$$n_{\text{BE}}(\epsilon) = \frac{1}{e^x - 1} = \frac{1}{e^{(\epsilon - \tilde{\mu})/k_{\text{B}}T} - 1}$$

Average Occupation Number

For a specific microstate with energy ϵ ,

- Classical particles (Maxwell-Boltzmann distribution)

$$n_{\text{MB}}(\epsilon) = e^{-(\epsilon - \tilde{\mu})/k_{\text{B}}T}$$

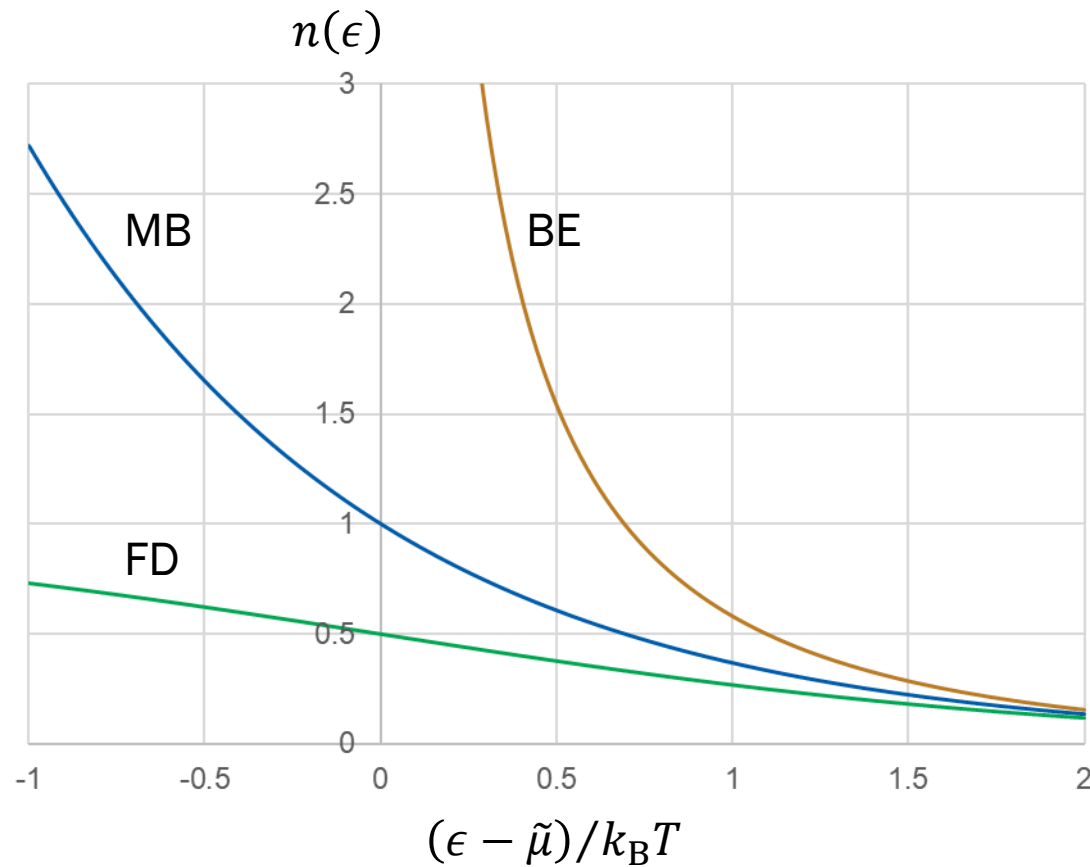
- Fermions (Fermi-Dirac distribution)

$$n_{\text{FD}}(\epsilon) = \frac{1}{e^{(\epsilon - \tilde{\mu})/k_{\text{B}}T} + 1}$$

- Bosons (Bose-Einstein distribution)

$$n_{\text{BE}}(\epsilon) = \frac{1}{e^{(\epsilon - \tilde{\mu})/k_{\text{B}}T} - 1}$$

Average Occupation Number



Chemical Potential

$$\tilde{\mu} = \left(\frac{\partial G}{\partial N} \right)_{T,p}$$

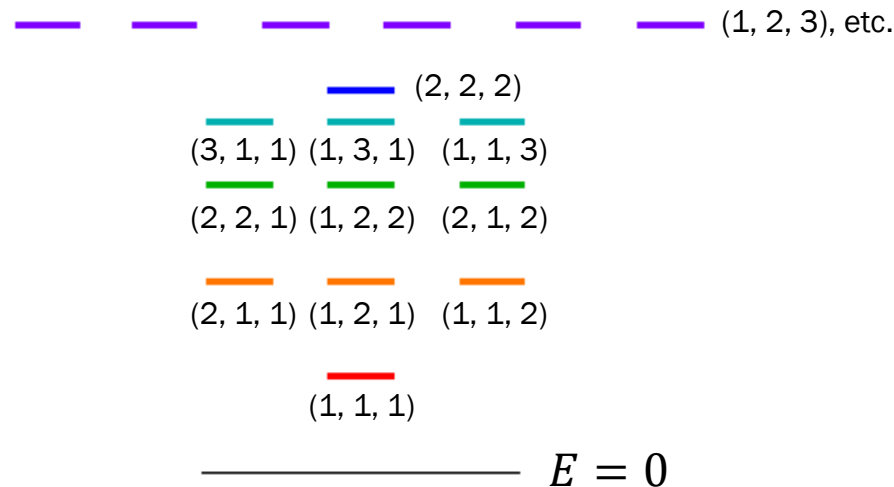
(Gibbs energy required to add one more particle)

Can we say something about the chemical potential?

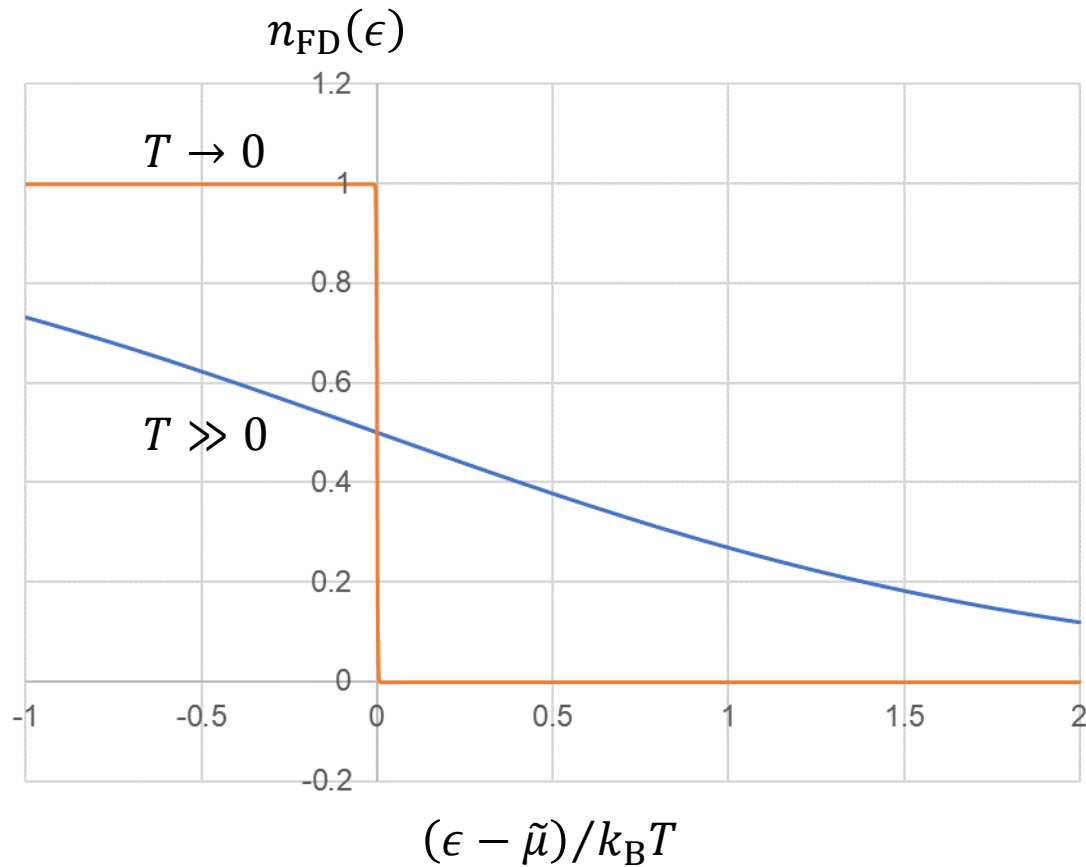
Quantum Particles in a 3D Box

- Microstate: specified by quantum numbers (n_x, n_y, n_z)
- Energy

$$E_{n_x, n_y, n_z} = \frac{h^2}{8ma^2} (n_x^2 + n_y^2 + n_z^2)$$



Chemical Potential of Fermion



Chemical Potential of Fermion

Fermi energy ϵ_F : the highest occupied energy level at 0 K

For $\epsilon > 0$,

$$\tilde{\mu} > 0$$

(fermions “feel” a repulsive potential)

Chemical Potential of Boson

$$n_{\text{BE}}(\epsilon) = \frac{1}{e^{(\epsilon - \tilde{\mu})/k_{\text{B}}T} - 1}$$

For $\epsilon > 0$ and $n_{\text{BE}}(\epsilon) > 0$,

$$\tilde{\mu} < 0$$

(bosons “feel” an attractive potential)